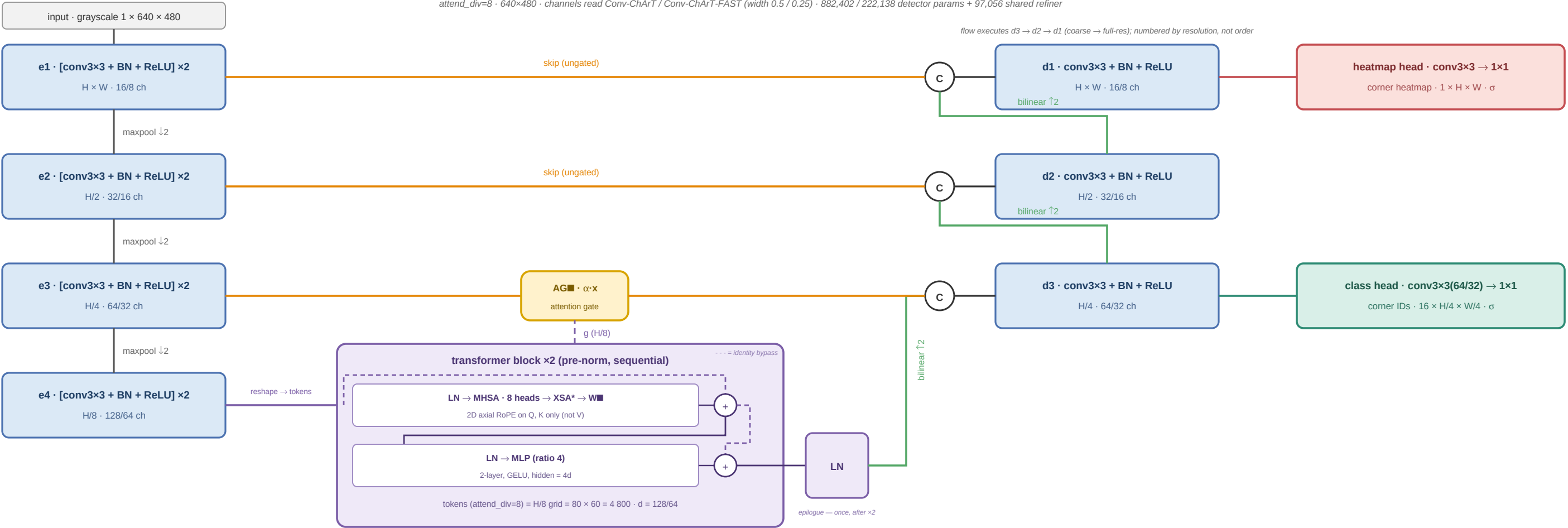


Conv-ChArT detector — hybrid convolutional–transformer architecture

attend_div=8 · 640×480 · channels read Conv-ChArT / Conv-ChArT-FAST (width 0.5 / 0.25) · 882,402 / 222,138 detector params + 97,056 shared refiner



Legend

- conv stage (blue): encoder ×2, decoder ×1 per box's own label
- transformer block: pre-norm MHSA+MLP, 2D axial RoPE on q,k · *XSA: FAST only, parameter-free
- attention gate: $\alpha = \sigma(\psi(\text{ReLU}(W_{\text{g}}x + W_{\text{g}}g)))$, pass-through init $\alpha_{\text{init}} \approx 0.95$
- maxpool ↓2
- bilinear ↑2
- skip connection
- gating signal g
- channel concat (C)
- residual add (+)
- output heads (heatmap · class); σ = sigmoid, losses on logits

Refiner (Stage 2, separate 97k-param network) — sub-pixel corner offset from a 24×24 crop



The 64×64 map covers only the central ±4 px capture window at 8× sub-pixel resolution (1/8-px cells); the surrounding 16-px ring is context via receptive fields, not answer space. Decode: soft-argmax over a 5×5 window around the peak → offset $d \in [-3.94, +3.94]$ px; refined corner = crop centre + d. Trained separately on jittered crops ($d \sim U$ over support).